Lesson 4: Project: Turtle Art Studio

Module 1 | 60 minutes | functions, input, menus | Grades 7–11 · ages 12–16

Name: _____ Class: _____ Date: _____

Plan your project and practise the key ideas.

1. Key words

Match each word to what it means. Write the letter.

a. menu ___ A. Running a part of your program to check it works before moving on.
b. interactive ___ B. A True/False variable, like drawing, that keeps a while loop going.
c. loop flag ___ C. A program that responds to what the user types.
d. testing ___ D. A program part that asks the user to choose and then acts on the choice.

2. Plan your program

Before you type it: Shape picker. Ask the user for a shape (square or triangle) and a size. Write a draw_triangle(size) function next to draw_square, and use if / elif / else to call the right one. For any other shape, print a message that starts with `Sorry`. Write the steps in plain English (pseudocode), then the Python you will use.

3. Predict the output: growing shapes

What does this program print? Code: def grow(size): ⏎ return size + 20 ⏎ ⏎ s = 40 ⏎ for i in range(3): ⏎ s = grow(s) ⏎ print(s)

a. A. 100
b. B. 60
c. C. 40
d. D. 120

4. Plan the studio

Put the parts of the Art Studio program in order Number them 1–6.

a. ___ Ask for the colour and size
b. ___ Ask for a shape (done ends the loop)
c. ___ Start a while loop
d. ___ Define a function for each shape
e. ___ import turtle and make the turtle t
f. ___ Move to a new spot and call the right shape function

5. Plan your program

Before you type it: Debug it! When the user types `Square` and `60`, this program should draw a closed square. Right now it draws nothing, and then it crashes. Fix both bugs. Write the steps in plain English (pseudocode), then the Python you will use.

6. My plan

Plan your project: Build an interactive drawing studio where the user picks shapes, colours and sizes, and every shape has its own function. Sketch it and list what each part will do.

7. Checklist

Tick each one when your project does it.

a. ☐ At least 3 shape functions, each with a size parameter
b. ☐ Asks for shape, colour and size with input()
c. ☐ A while loop that keeps drawing until the user types done
d. ☐ A friendly Sorry message for unknown shapes
e. ☐ Shapes moved apart and counted, with a final message
f. ☐ A title screen and a “Made by” comment

8. Two stars and a wish

Ask a friend to test your project. Write two things they liked and one idea to improve it.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can write a separate function for each shape with a size parameter.
☐ I can build a menu with input and if / elif / else.
☐ I can keep a program running with a while loop until the user types done.
☐ I can test each function on its own before joining them together.